CRL Seminar Defense Guide: All 100 Questions & Answers

Offline Edge-AI-Based 2D/3D Motion Tracking and Activity Classification

Part I: Introduction, Motivation & Scope (Q1–Q10)

Q1 Primary motivation behind an offline Edge-AI framework over cloud?

Traditional cloud systems introduce latency, consume excess energy for wireless transmission, and pose data-privacy risks. Local processing ensures real-time response, zero cloud dependence, and robust privacy.

Q2 Why is standard GPS inadequate for indoor positioning?

GPS signals cannot reliably penetrate buildings, causing multipath propagation, signal attenuation, and severe vertical/horizontal drift errors (drifting by several meters vertically).

Q3 Does this paper claim to present a complete, commercial indoor navigation solution?

No. It demonstrates a lightweight embedded framework for motion-related information and activity classification as a foundation for future GPS-denied indoor positioning.

Q4 What are the target applications for this system?

Personal health monitoring, indoor navigation in GPS-denied environments, first-responder tracking, and secure industrial/defense security systems.

Q5 Why is this work relevant to Bharat Electronics Limited (BEL / CRL)?

It demonstrates indigenous development of low-power, secure, standalone tactical navigation tools operating without external infrastructure—critical for defense and secure environments.

Q6 How does this system handle data privacy?

All sensor acquisition, signal processing, feature extraction, and ML inference happen locally on microcontroller SRAM/Flash; no raw user movement data ever leaves the device.

Q7 What distinguishes this from standard smartphone pedometer apps?

Smartphones rely on heavy OSs and cloud sync. This framework runs deterministically on a bare-metal/lightweight MicroPython environment on a microcontroller with minimal memory overhead (<15 KB RAM).

Q8 What does "Edge-AI" mean in this context?

Executing both the signal conditioning pipeline and the machine-learning classification model directly on the edge hardware (the microcontroller) rather than offloading to a remote server.

Q9 What core metrics are tracked simultaneously?

Step count, heading direction (8 sectors), cumulative distance, vertical altitude/floor changes, and physical activity state (stationary, walking, running).

Q10 Why choose MicroPython instead of C or C++?

MicroPython enables rapid prototyping, readable logic, and straightforward translation of ML conditional rules while maintaining acceptable execution speed on the high-performance Arm Cortex-M7 core.

Part II: Hardware Architecture & Crossover MCU (Q11–Q20)

Q11 Specific microcontroller evaluation kit used?

The NXP MIMXRT1064 evaluation kit, featuring the i.MX RT1064 crossover processor.

Q12 Why is the i.MX RT1064 classified as a "crossover processor"?

It combines the high processing performance and deterministic behavior of a microcontroller with the high-integration peripheral set typically found in microprocessors.

Q13 Core clock speed, Flash, and SRAM specifications?

Operates at up to 600 MHz on an Arm Cortex-M7 core, providing 4 MB of on-chip Flash memory and 1 MB of SRAM.

Q14 How are sensors interfaced with the processor?

Interfaced through the hardware Inter-Integrated Circuit (I^2C) interface operating at a standard speed of 100 kHz.

Q15 What specific sensor modules are connected?

STMicroelectronics LSM303DLHC module (3-axis accelerometer & magnetometer) and TDK InvenSense ICP-10111 high-accuracy barometric pressure sensor.

Q16 What sensing capability does the LSM303DLHC provide?

It provides 6 degrees of freedom (DoF)—3 axes for linear acceleration and 3 axes for magnetic field measurement.

Q17 What makes the ICP-10111 barometer suitable?

It is an ultra-low noise, waterproof sensor capable of resolving minute, localized changes in atmospheric pressure down to centimeter-scale altitude variations.

Part III: Sensor Configuration, Calibration & Preprocessing (Q21–Q40)

Q21 Sampling frequencies for accelerometer and magnetometer?
Accelerometer is configured at 100 Hz (high-resolution), while the magnetometer is set to continuous-conversion mode at 30 Hz.

Q22 Why is sensor calibration necessary before capture?
To compensate for baseline mounting tilt, gravity-related offsets on the accelerometer, and hard-iron distortions on the magnetometer.

Q23 How is accelerometer baseline bias calculated?
By averaging 100 stationary samples during initialization using: $Bias_q = (1 / N) \Sigma Q_i$ where $N = 100$.

Q24 What causes hard-iron distortions on a magnetometer?
Nearby permanent magnets, magnetized metallic components, or electronic chassis parts creating a constant offset magnetic field.

Q25 How is magnetometer hard-iron compensation performed?
Through an interactive rotational sweep where min and max magnetic field measurements across axes are recorded to calculate horizontal offsets: $V_{offset,x} = (\max(M_x) + \min(M_x)) / 2$.

Q26 Difference between hard-iron and soft-iron distortions?
Hard-iron adds a constant offset vector, whereas soft-iron warps the magnetic field shape. This paper compensates for hard-iron offsets.

Q27 How are calculated offsets applied during live tracking?
Computed $V_{offset,x}$ and $V_{offset,y}$ values are subtracted from subsequent raw magnetometer readings before heading calculation.

Q28 Does the system perform 3D tilt compensation on the compass?
No. The heading implementation is based on calibrated horizontal measurements and is not a tilt-compensated 3D compass solution (future work).

Q29 Why does failure to calibrate lead to dead-reckoning failure?
Uncompensated biases accumulate rapidly during integration, causing heading vectors to skew and distance calculations to drift severely.

Q30 How does the system handle air drafts in barometric readings?
It applies a centered moving-average filter ($N = 3$) to flatten ambient pressure fluctuations and air drafts, preserving a smooth vertical profile.

Part IV: Signal Processing, Step Detection & Distance (Q31–Q40)

Q31 How does the system detect an individual step?
By computing differential acceleration magnitude (ΔM) between consecutive samples and checking if it crosses threshold $TH_{step} = 180$.

Q32 What is the formula for differential acceleration magnitude (ΔM)?
$\Delta M = \sqrt{((x_t - x_{t-1})^2 + (y_t - y_{t-1})^2 + (z_t - z_{t-1})^2)}$.

Q33 What is the numerical value assigned to the step threshold?
$TH_{step} = 180$ (in raw sensor units).

Q34 Purpose of the temporal cooldown mechanism ($T_{cooldown}$)?
Applied after a step is registered ($T_{cooldown} = 400$ ms) to suppress multiple false triggers caused by a single physical foot-strike impact.

Q35 How does foot-strike mechanical recoil affect detection?
Foot impact creates short-duration mechanical recoil or back-and-forth motion that can trigger false reverse-direction indications or duplicate step spikes.

Q36 How is distance calculated for each step initially?
A fixed step length (L_O) is assigned as a baseline to each detected step ($L(k) = L_O$).

Q37 Why is a moving-average operation applied to step distances?
To smooth out variations in individual stride lengths and yield a more consistent cumulative distance estimate.

Q38 What is the formula for moving-average step distance?
$L_{MA}(k) = (1 / N) \Sigma L(k - i)$ across window N.

Q39 Limitations of using a fixed step-length baseline?
It cannot adapt to changes in walking speed, terrain slope, fatigue, or individual user gait characteristics, introducing residual systematic error.

Q40 How is horizontal heading angle calculated?
$\theta = (\text{atan2}(M_y - V_{offset,y}, M_x - V_{offset,x}) \times 180 / \pi) \bmod 360$.

Part V: Heading, Barometric Altitude & 3D Mapping (Q41–Q60)

Q41 Into how many directional sectors is heading mapped?
Eight directional sectors: North, North-East, East, South-East, South, South-West, West, and North-West.

Q42 How does the barometric sensor assist in multi-floor navigation?
By capturing micro-variations in atmospheric pressure corresponding to height changes, allowing differentiation between stacked building levels.

Q43 What baseline atmospheric pressure was recorded at origin?
910.48 hPa at the ground floor origin point.

Q44 How does atmospheric pressure change when moving upstairs?
Atmospheric pressure decreases as altitude increases (roughly 0.12 hPa difference observed between ground and upper floors).

Q45 Vertical resolution achieved by incorporating ICP-10111?
Centimeter-scale vertical resolution, effectively eliminating multi-meter vertical GPS drift.

Q46 How does the system fuse 2D planar movement with barometer data?
The X–Y dead-reckoning trajectory is combined with the Z-axis barometric elevation metric to form a true 3D indoor trajectory.

Q47 What specific waypoint event occurs at Pressure 910.36 hPa?
"Started stairs up" (Elevation 0.74 m, Heading South 180°).

Q48 What happens at waypoint 3 in the experimental trial?
Turning halfway on stairs / turning to 1st floor (Heading West 270°, Pressure 910.20 hPa, Elevation 2.66 m).

Q49 What is the final destination state (waypoint 5)?
Reached 1st floor lab destination (Stationary, Pressure 910.01 hPa, Elevation 4.48 m, Distance 26.0 m).

Q50 How does Fig 2 demonstrate successful sensor fusion?
It plots a clear 3D trajectory showing horizontal turns combined with vertical elevation steps as the user ascends stairs and enters a stable floor.

Part VI: TinyML Activity Classification (Q51–Q60)

Q51 What machine learning model is used for activity classification?
A Scikit-Learn Decision Tree classifier trained offline on a PC.

Q52 What three physical activity states are classified?
Stationary, Walking, and Running.

Q53 How many samples were in the labeled training dataset?
1,500 samples collected using sensor-derived motion features and heading information.

Q54 What is the maximum depth of the trained Decision Tree?
A maximum depth of three.

Q55 How is the model deployed without a heavy ML runtime?
Translated offline into explicit deterministic conditional statements (if-else rules) embedded directly in the MicroPython application.

Q56 Why is translating a Decision Tree into conditional rules advantageous?
It eliminates the need for heavy ML inference engines (like TFLite), drastically reducing RAM/Flash overhead and execution latency.

Q57 What is the classification latency on the Cortex-M7 core?
Less than 0.1 ms per frame.

Q58 What is the memory overhead of the deployed classification logic?
Below 15 KB of RAM.

Q59 Why weren't Deep Neural Networks (DNNs) or CNNs used instead?
DNNs/CNNs require significant memory and floating-point compute overhead, whereas a shallow Decision Tree achieves high accuracy with minimal resource utilization.

Q60 How does the classifier handle ambiguous transitional states?
With a shallow depth-3 tree and distinct feature thresholds, intermediate states map cleanly, and framing logic smooths rapid transitions.

Part VII: Experimental Results & Performance Metrics (Q61–Q80)

Q61 Step-counting accuracy for Slow Walking (100 actual steps)?
96.0% (96 detected steps, estimated distance 48.0 m).

Q62 Step-counting accuracy for Normal Walking (200 actual steps)?
97.5% (195 detected steps, estimated distance 97.5 m).

Q63 Step-counting accuracy for Brisk Walking (150 actual steps)?
99.3% (149 detected steps, estimated distance 74.5 m).

Q64 Step-counting accuracy for Running/Jogging (120 actual steps)?
96.7% (116 detected steps, estimated distance 58.0 m).

Q65 Overall offline activity-classification accuracy?
98.2% overall accuracy.

Q66 Confusion matrix: Stationary samples correctly predicted?
495 correctly predicted out of 500 actual stationary samples (only 5 misclassified as walking).

Q67 Confusion matrix: Walking samples misclassified as Running?
8 samples misclassified as running (out of 500 actual walking; 488 correct, 4 stationary).

Q68 Confusion matrix performance for Running class?
490 correctly predicted out of 500 running samples, with 10 misclassified as walking and 0 as stationary.

Q69 Distance drift error reduction with TinyML?
Distance drift error dropped from 12.4%–18.2% down to 2.8%–4.1%.

Q70 Horizontal position error improvement?
Improved from 3.45 m down to 0.68 m over a test trajectory.

Q71 Floor detection accuracy comparison?
Traditional non-AI methods achieved 76.2%, whereas the proposed framework achieved 97.8%.

Q72 False step trigger rate comparison?
Dropped from 18.5% (traditional) to under 1.5% (proposed TinyML framework).

Q73 Stride model dynamics difference?
Traditional methods use a fixed stride model (0.50 m), whereas the proposed framework incorporates adaptive/moving-average step dynamics.

Q74 Why do fixed-threshold PDR systems have high false triggers?
Minor body shifts, desk bumps, or handling artifacts easily cross static acceleration thresholds unless filtered by cooldown timers.

Q75 How does activity state classification improve step counting?
If the classifier determines the user is "Stationary," the system suspends step-counting logic, preventing false steps while standing still.

Q76 Role of moving-average filter in eliminating air-draft spikes?
Averaging pressure readings over a rolling window (N = 3) smooths out sudden pressure anomalies caused by doors opening or HVAC drafts.

Q77 Total traversed distance recorded in the trial table?
26.0 meters from origin to destination.

Q78 Specific heading recorded at destination waypoint (Waypoint 5)?
North-East (NE, 45°).

Q79 Significance of 26 m distance across 5 waypoints?
Demonstrates continuous tracking from building entrance, up stairs, across landing, and into lab.

Q80 Why combine accelerometer, magnetometer, and barometer?
Inertial sensors drift vertically over time; adding a barometer locks down absolute floor/altitude tracking, constraining 3D drift.

Part VIII: Limitations, Future Work & Key Takeaways (Q81–Q100)

Q81 Principal limitation of current directional implementation?
Susceptibility to environmental magnetic interference, nearby metallic structures, and lack of 3D tilt compensation.

Q82 How do nearby metallic structures affect magnetometers?
They introduce soft-iron and hard-iron distortions that warp local magnetic field lines, leading to heading errors if uncompensated.

Q83 Why does the lack of 3D tilt compensation matter?
If the sensor module is tilted away from the horizontal plane during movement, 2D compass calculations project vertical magnetic components onto horizontal axes.

Q84 Does the fixed step-length baseline account for user height?
No. The current baseline uses a generalized fixed step length; user-specific calibration is noted as future work.

Q85 Can this system track sideways or backward walking?
The differential acceleration and 8-sector heading model are optimized for forward progression; sideways/backward requires multi-axis velocity profiling.

Q86 What happens to trajectory if device is in hand vs pocket?
Orientation changes alter axis mapping. The current system assumes a fixed mounting orientation; arbitrary placement requires transformation.

Q87 How does battery drain affect sensor sampling over long durations?
Continuous 100 Hz polling over MicroPython consumes power; power-gating or interrupt-driven wake-up would be needed for ultra-long deployments.

Q88 Limitations associated with using MicroPython?
Introduces garbage collection pauses and interpreter overhead compared to compiled C/C++, though the Cortex-M7 easily absorbs this.

Q89 How many training subjects were used for the 1500 samples?
The paper collects 1500 labeled motion/heading samples but does not detail a large demographic spread, a common limitation in prototypes.

Q90 Does the barometer get confused by weather pressure changes?
Over hours, weather pressure shifts occur. For indoor floor-change tracking over short operational windows, HVAC pressure differences dominate.

Q91 Primary avenues for future work outlined by authors?
Adaptive step-length estimation, multi-axis tilt compensation, improved magnetometer calibration, and multi-sensor fusion for GPS-denied indoor positioning.

Q92 How will adaptive step-length estimation improve the model?
It will dynamically adjust stride length based on walking speed, stride frequency, and activity state.

Q93 What sensor additions are planned for future positioning?
Integrating a gyroscope for rotational tracking alongside accelerometer, magnetometer, and barometer to compute full X–Y trajectory.

Q94 How can magnetometer calibration be enhanced in future?
By implementing dynamic background ellipsoid fitting algorithms (soft-iron matrix calibration) rather than static min-max sweeps.

Q95 Can this framework be extended to underground mines or tunnels?
Yes! Because it operates completely offline without cloud or GPS dependence, it is ideally suited for underground mines, tunnels, and tactical environments.

Q96 Ultimate takeaway message for seminar judges?
Lightweight motion analytics and TinyML inference can run locally on an affordable Cortex-M7 MCU with high accuracy (>96% step, 98.2% activity) and zero cloud dependence.

Q97 How does this contribute to "Edge Intelligence for Smart Sensing"?
It provides a concrete, resource-efficient blueprint for deploying machine learning and sensor fusion at the extreme edge, minimizing latency and security vulnerabilities.

Q98 What tool was used to train the machine learning model?
Python with the Scikit-Learn library running on an offline PC.

Q99 Why is MicroPython + conditional rules novel for constrained MCUs?
It bridges high-level data science model training with bare-metal interpreter execution without needing complex C++ library ports or heavy ML runtimes.

Q100 Ultimate elevator pitch response for judges?
"We achieve 98.2% activity classification and >96% step accuracy locally on a microcontroller with <15 KB RAM and <0.1 ms latency, eliminating cloud dependency and reducing position drift from 12% down to under 4%."